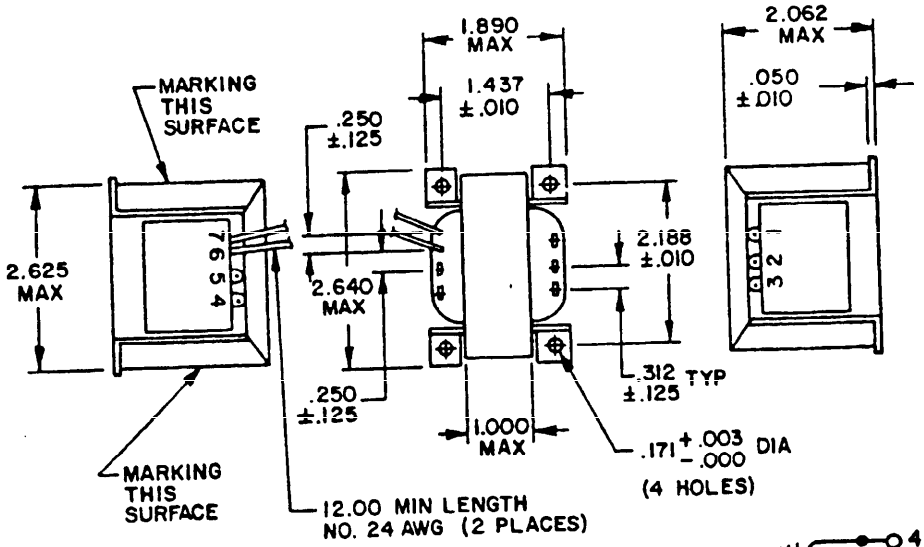
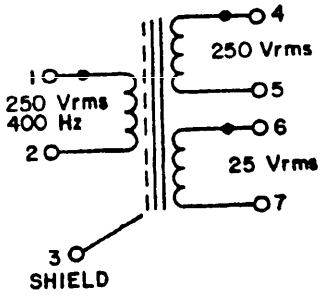
MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, 50 VA, TF5T03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification and the latest issue of MIL-T-27.





WORKING VOLTAGE: 250 V (EACH WINDING)
ALTITUDE: 70,000 FEET MAXIMUM

CIRCUIT DIAGRAM AND MARKING

Inches	mm	Inches	mm
.003	0.08	1.437	36.50
.010	0.25	1.890	48.01
.050	1.27	2.062	52.37
.125	3.18	2.188	55.58
.171	4.34	2.625	66.68
.250	6.35	2.640	67.06
.312	7.92	12.000	304.80
1.000	25.40		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the sides of the case.

FIGURE 1. Dimensions and configuration.

(A) denotes changes

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FSC 5950
AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the windings.)

Electrical ratings:

Primary: (1-2): 50 volt-amperes, 250 volts rms, 400 hertz.

Secondary:

(4-5): 250 volts rms, 0.2 ampere rms.

(6-7): 25 volts rms, 0.01 ampere rms.

Working voltage:

(1-2): 250 volts rms

(4-5), (6-7): 250 volts rms.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: (5) Solder lugs (copper, tin-plated), and (2) insulated wire leads (No. 24 AWG, type E per MIL-W-16878).

Height: Lugs (.375 inch (9.53 mm) maximum), leads (12 inches (304.8 mm) minimum length).

Weight: 1.25 pounds maximum.

Operating temperature range: -55°C to +170°C.

Ⓐ Altitude: 70,000 feet maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage (each winding):

At sea level: 1,000 volts rms.

At reduced barometric pressure: 312.5 volts rms.

Electrical characteristics:

No load: With 250 volts rms and 400 hertz in (1-2), and

Ⓐ Current in (1-2) at 75 milliamperes rms maximum.

Power in (1-2): 1.0 watt maximum.

Voltage across (4-5): 250 volts rms ±10 percent.

Voltage across (6-7): 25 volts rms ±10 percent.

Ⓐ Rated load: With 250 volts rms and 400 hertz (1-2) and rated current in all secondaries:

Voltage across (4-5): 250 volts rms +10 percent, -20 percent.

Voltage across (6-7): 25 volts rms +10 percent, -20 percent.

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DC resistance:

(1-2): 6.0 ohms ±20 percent.

(4-5): 7.0 ohms ±20 percent.

(6-7): 7.0 ohms ±20 percent.

Electrostatic shielding: Voltage ratio 5 to 1 at 400 hertz.

Polarity: Additive, with terminals 2 and 4, and 5 and 6 connected.

Temperature rise: 75°C with 250 volts rms, 400 hertz across (1-2) at an ambient temperature of 95°C, full load on terminals (4-5) and (6-7).

Vibration, high frequency: MIL-STD-202, method 204, test condition D.

Salt spray (corrosion): MIL-STD-202, method 101, test condition B.

Part number: M27/212-01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0666-02)

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